

SHIZHE “ANDY” ZHANG

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Berkeley, California, USA

Education

University of California, Berkeley | **GPA:** 4.0/4.0

08.2025 - 12.2026

M.A. in Statistics & Data Science

Advanced topics in Neural Networks, NLP, Statistical Computing, Statistical Learning Theory, Optimization Models in Engineering

Zhejiang University | **GPA:** 3.96/4.0

08.2021 - 06.2025

B.S. in Economics (*Advanced Class*, Selected from top 10%)

Game Theory, Incentive Theory, Econometrics, Statistical Software, Time Series, Quantitative Investment, Big Data Analysis

Minor (Mathematics): Real Analysis, Stochastic Process, Numerical Analysis, PDE, Statistical Learning, Data Structure

Honor: First Prize in National College Mathematics Competition, Certificate of the Chu Kochen Honors Program (top 1.3%), First Prize Scholarship for Outstanding Economics Students (top10%), Outstanding Graduates, College-level Scholarship

Skills

- **Programming:** Python, TypeScript, R, SQL (MySQL, PostgreSQL), Matlab, Shell Scripting (Bash, Zsh)
- **Tools & Software:** Git, Docker, FastAPI, Next.js/React, Cloudflare Workers, Cloud (AWS, GCP), Unix, Latex
- **Technical Skills:** Machine Learning, Deep Learning, Feature Engineering, Causal Inference, Model Evaluation, Optimization, NLP, LLM/VLM Agents, RAG, MCP, vLLM, Structured Decoding, LLM-as-Judge, Prompt Engineering

Internships

Monolithic Power Systems, AI Developer

05.2026 - 08.2026

- Built a layout-preserving translation agent for PDF/DOCX/PPTX. Designed a multi-level memory: a short-term window for local coherence, retrieved exemplars from past translations, and proper-noun records pinning terms document-wide. Per-language-pair LoRA adapters hot-swap on self-hosted vLLM, raising COMET ~ 3 points across all pairs.
- Designed an in-house SEC EDGAR MCP server for the finance team with edge caching and grounded, cited figures.
- Engineered a slide-generation agent compiling typed actions (charts, tables, Mermaid) via python-pptx. One action registry generates the model's schema, runtime validation, and execution dispatch. Every edit appends to an immutable log, giving decks undo and branching version history; content quality rose 20% on a PPTEval-style rubric.

SDIC Securities, Quantitative Researcher

10.2023 - 03.2024

- Generated event-driven strategies from the A-share Billboard Stocks (*Dragon-Tiger List*).
- Built predictive features from raw trade data, including net inflow strength, institutional concentration, and order imbalance.
- Trained a LightGBM to predict short-term price movements. Top-quintile portfolio with a monthly excess return of 2.3% over the market benchmark.
- Constructed a capital-activity factor by modeling synergistic trading behavior using brokerage network analysis. Deployed the factor in a long-short strategy (CSI 300 vs. CSI 2000), boosting signal precision by 14% over heuristic benchmarks.

Projects

Cost-Aware Clarification in Multimodal Reference Games 🍌

04.2026 - 05.2026

- Derived a training-free expected-utility rule for when a vision-language listener should ask versus commit (commit iff top posterior $\geq 1 - c$); built the speaker/listener framework, evaluation pipeline, and metrics suite.
- Ran 28,800 zero-shot trials across 8 speaker and 6 listener strategies (Gemini-2.0-Flash) over 600 rendered scenes, scoring cost-penalized accuracy against clarification rate.
- Showed Chain-of-Thought *lowers* cost-penalized accuracy—correct on 76.5% of scenes yet asking on 62.9%—and that multi-turn clarification halves accuracy on the scenes it flags, isolating miscalibrated uncertainty rather than reasoning ability as the bottleneck.

Rag-Driven Company Search System (Industry Partner: Pyramyd) 🍌

10.2025 - 12.2025

- Built RAG-based discovery system over 100K+ reviews, integrating keyword search, embeddings, and FAISS retriever.
- Conducted sensitivity and counterfactual ablation analyses to validate robustness, demonstrating 85–90% top- k ranking stability and 20–35% Recall@K degradation when key components were removed.
- Implemented a modular experimentation pipeline spanning data parsing, chunking, indexing, caching, grounding, and evaluation, enabling systematic ablations across 10+ retrieval and reranking variants while avoiding full reindexing for 80% of experiments.
- Delivered strong system performance using hybrid candidate fusion and metadata reranking, improving retrieval relevance by 30-50% (Recall@K / NDCG@10), achieving real-time ($< 1s$) retrieval, and reducing average LLM prompt size by 40%.

Robust Portfolio Optimization of CNE5 Factors 🍌

11.2024 - 12.2024

- Standardized 100+ raw style/industry factors, applied sparse PCA factor selection to derive interpretable and stable factor structures, and estimated factor covariance matrices under high dimensionality.
- Optimized portfolios using mean-variance and CVaR minimization under sector-neutrality and turnover constraints.
- Evaluated robustness via 10,000+ Monte Carlo stress scenarios and block bootstrap resampling.
- Benchmarked Barra-style linear risk models against ML-enhanced covariance estimation, including Ledoit-Wolf shrinkage and Graphical Lasso for high-dimensional stability, improving Sharpe ratio by 11% and reducing max drawdown by 6%.